Digital Economy and Sustainable Development

REVIEW

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Boosting the intra-African digital trade in the AfCFTA context: does regulatory framework matter?

Manfred Kouty[1*]

Abstract

Digital trade is one of the promising areas for the African Continental Free Trade Area (AfCFTA) and a factor of success of the African Union Digital Transformation Strategy (DTS) for Africa (2020–2030). However, the benefits of digital trade are not automatic and require an adequate regulatory environment. The objective of this study is to show how an adequate regulatory framework can boost the intra-African digital trade in the AfCFTA context. Our descriptive analysis shows that digital trade has increased across all regions in the world particularly in Africa since the beginning of the covid-19 pandemic. As a share of digitally deliverable services in total export services, digital trade represents 35% of total export services in Africa. Digital trade expansion faces regulatory barriers and the lack of a harmonized regulatory framework. Our econometrics results shows that digital trade and digital trade regulatory barriers (RDTII) are strongly negatively correlated in Africa. This result establishes the need to improve digital trade regulations framework in Africa.

Keywords Digital trade, e-commerce, Regulatory barriers, African Continental Free Trade Area

JEL F13, F15, K20, O55

1 Introduction

Since the 18th African Union Summit held in Addis Ababa in January 2012, the Heads of States and Governments had endorsed a declaration on "Boosting Intra-African Trade (African Union 2013). In this regards, they decided the establishment of the African Continental Free Trade Area (AfCFTA). The agreement establishing the AfCFTA was signed by 44 countries at a summit of the African Union held in Kigali (Rwanda) in March 21, 2018. Considered as a keystone initiative of AU's Agenda 2063, the AfCFTA was officially entered into force on 3 may 2019 and its operationalization is effective since the first of January 2021.[1]

Once operational, the AfCFTA is expected to become the world's largest free trade area by the number of participating countries, representing a market of more than 1.3 billion people with a combined GDP exceeding US$2.5 trillion (World Bank 2020; 2022). Initially, AfCFTA covers trade in goods, trade in services, investment, intellectual property and competition policy. The negotiations on the AfCFTA protocol have been divided into three phases. The first phase of the negotiations concern trade in goods and services. The second phase of the negotiations concern investment, intellectual property and competition policy. The last phase of negotiations

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[1] See figure A1 in appendix for the members of AfCTA. As at May 2022, 43 of the 54 signatories (80%) have deposited their instruments of AfCFTA ratification.

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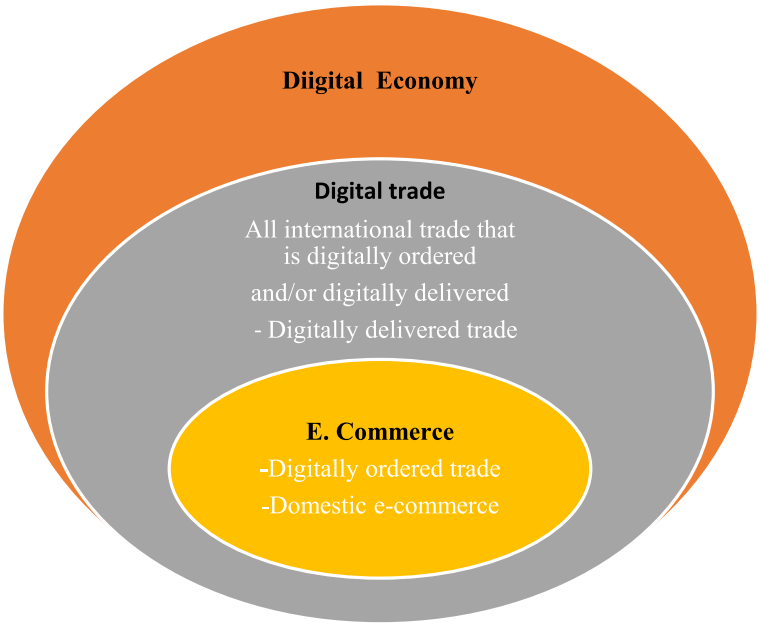


Fig. 1 Digital trade and e-commerce

concerns women and youth, and e-commerce. Absent in the AfCFTA agreement, e-commerce has been included in the AfCFTA through a decision[2] of the African Union Heads of State and Government held in February 2020 in Addis Ababa. The decision urges "Member States to critically review approaches that are being made to them by bilateral partners to enter into bilateral e-Commerce legal instruments with them in order to ensure that Africa is able to negotiate and implement an AfCFTA Protocol on e-Commerce where Africa has full authority on all aspects of e-commerce such as data and products being traded under e-commerce, and to promote the emergence of African owned e-Commerce platforms at national, regional and continental levels as part of our preparations for the negotiation of an AfCFTA Protocol on e-Commerce".

Despite being placed in the third phase of the negotiations, digital trade is one of the promising areas for the AfCFTA and a factor of success of the African Union Digital Transformation Strategy (DTS) for Africa (2020–2030). The implementation of the AfCFTA, coupled with the DTS, is expected to provide the creation of a continental digital single market. Although existing data show increase in digital trade in Africa, the volume of digital trade remains weak compared to other regions of the world. Digital trade approached, respectively, 35%, 68% and 72% of total services exports in Africa, Europe and North America (UNCTAD).[3] This low performance is the results of many factors, including the lack of infrastructures and enabling legal, policy and institutional frameworks on digital trade. The recent economic literature emphasizes that, regulation framework is one of the important challenges to build a digital trade in the context of AfCFTA. Numerous studies (Ogo 2020; Banga et al.2021; World Bank 2022) have investigated such factors using narrative approach and have put relatively less emphasis on the regulatory factors by using descriptive and econometric approaches. The objective of this study is to show that an adequate regulation framework is an essential condition for boosting intra-African digital trade in the AfCFTA context. The rest of the paper proceeds as follows: the next section presents the literature review while the Sect. 3 is devoted to the empirical analysis. Section 4 concludes.

2 Literature review

2.1 Understanding the concept of digital trade

Before the appearance of the concept of digital trade, the concept of e-commerce was used in the literature to refer selling and delivering of goods and services using electronic means and computer networks (Ebrahimi Darsinouei and Kaukab 2017). However, there is

[2] https://au.int/sites/default/files/decisions/38180-assembly_au_dec_749-795_xxxiii_e.pdf.

[3] The data are available on UNCTAD's website: https://unctadstat.unctad.org/datacentre/

no universally accepted and precise definition of e-commerce. The two most used definitions of e-commerce are those suggested by the OECD[4] and the WTO.[5] According to the OECD (2011) definition, e-commerce refers to the sale or purchase of goods or services, conducted over computer networks by methods specifically designed for the purpose of receiving or placing of orders. This definition is reductive insofar as it requires that the order should be made online for the transaction to be considered as e-commerce. The WTO has established a more general definition of e-commerce, which is understood to mean "the production, distribution, marketing, sale or delivery of goods and services by electronic means" (WTO 1998). In this definition, a transaction can be classified as e-commerce, even if some part of it is not being done by electronic means.

Nowadays, flows of data are exchange between countries around the world. Thereby, data flows are not only a means of production, but also an asset that can themselves be traded, and a means through which Global Value Chains (GVCs) are organized and services delivered (López González and Jouanjean 2017). This aspect goes beyond the scope of e-commerce and has favored the emergence of the concept of digital trade. Digital trade has a broader scope than e-commerce and captures not just the sale of consumer products on the internet and the supply of online services, but also data flows that enable global value chains, services that enable smart manufacturing, and myriad other platforms and applications (USTR 2017). It is therefore clear that digital trade includes e-commerce, and constitute an important part of the growing digital economy (Fig. 1).

But what is digital trade? However, there is not yet a universally recognized definition of digital trade due to its complexity (TRALAC, 2023). This is mainly due to the multidimensional nature of the transactions but also to the multitude of actors involved. Depending on the nature of the transaction, the best known and most current definition, is that proposed by the IMF, OECD, UN and WTO((2023) which define digital trade as "all trade (exports/imports) that is digitally ordered and/or delivered". In this definition, digitally ordered trade is refers to the international sale or purchase of a good or service, conducted over computer networks by methods specifically designed for the purpose of receiving or placing orders. In view of this definition, digitally ordered trade is synonymous of international e-commerce (OECD et al.2019). On the other hand, digitally delivered trade refers to the international transactions that are delivered remotely over computer networks (IMF et al. 2023).

By considering the actors involved in the transaction, we distinguish several types of digital trade models: Business to Business(B2B), Business to Consumer(B2C), Consumer to Consumer (C2C), Consumer to Business (C2B), Consumer to Government (C2G), Business to Government (B2G), Government toBusiness (G2B) and Government to Consumer(G2C) (Isaac and Volle 2008, Jaller et al. 2020).

2.2 Digital trade in AfCFTA agreement

As mentioned in the introduction, e-commerce has been included in the AfCFTA through a decision of the AU Heads of State and Government during the Ordinary Session held in Ethiopia in February 2020. It is important to note that the term "e-commerce" is used instead of digital trade in that decision. In May 2021, the Council of Ministers of AU established the Committee[6] to coordinate and facilitate the negotiations of the Protocol on Digital Trade under the AfCFTA. The protocol in Digital Trade was scheduled for Phase 3 and expected to kick-in immediately after conclusion of Phase 2 negotiations, which include competition policy, intellectual property rights and investment protocol.[7] At the end of the negotiation, the Protocol, upon adopted and entry into force, would form an integral part of the AfCFTA Agreement.[8]

To date, negotiations of AfCFTA protocol on digital trade have not yet started. Given the complexity of digital trade combine with the absence of global agreement on digital trade to use as a benchmark, some important issues have to be considered in the formulation of an agenda for the AfCFTA protocol on digital trade. On the basis of the literature (Ebrahimi Darsinouei and Kaukab 2017; López González and Jouanjean 2017; Ogo 2020), negotiations must pay attention on two categories of issues: the conceptual and enabling issues (Fig. 2).

The conceptual issues are very important regarding the absence of universal definition of digital trade. A precise definition of digital trade in the AfCFTA and all issues is central to the direction of the protocol. The quality of provisions to put in place depend on a better understanding of the concept of digital trade.

In order to promote the adoption and use of the technologies that facilitate electronic transactions in the context of digital trade, discussions must be focused on a range of regulatory principles with a view to achieving a common understanding of what the key issues for regulation might be (Nemoto and López González 2021). The

[4] The Organization for Economic Cooperation and Development.

[5] The World Trade Organization.

[6] The Committee on Digital trade.

[7] See the Table 1 in annex for more information about the AfCFTA phases of negotiations.

[8] For more informations, go the AfCFTA website at https://au-afcfta.org/trade-areas/digital-trade/.

Table 1 Data and sources

Variables	Definitions	Sources
Digitaltrade	Bilateral digitally deliverable trade (as share of total services export) Digitally deverable trade is identified by using the WTO's (2016) classification of digitisable goods (films, software, videogames, printed matter, sound & music)	UN Comtrade (WITS and WTO's (2016) HS six-digit classification of products) https://wits.worldbank.org/WITS/WITS/AdvanceQuery/RawTradeData/QueryDefinition.aspx?Page=RawTradeData
Distance	Great circle distance between the two partners	CEPII
Contiguity	Dummy variable equal to unity if countries i and j share a common land border	CEPII
Common Official Language	Dummy variable equal to unity if countries i and j share a common official language	CEPII
Common currency	Dummy variable equal to one if the two countries share the same currency	CEPII
Commun colonizer	Dummy variable equal to unity if economies i and j were colonised by the same power	CEPII
RTA	Dummy variable equal to unity if countries i and j are members of the same RTA	WTO
RDTII	Bilateral Regional Digital Trade Integration Index, as a proxy for digital trade regulatory environment. Its the geometric average of the two partners RDTII (Indicators scores range from **0** (reveals that the policy environment is conducive for digital trade) to **1** (highlights a policy framework that can hamper digital trade)	United Nations Economic Commission for Africa (UNECA). Available at: https://repository.unescap.org/rest/bitstreams/04cbf0ef-1c51-4e23-9d5c-6a7147e3e276/retrieve

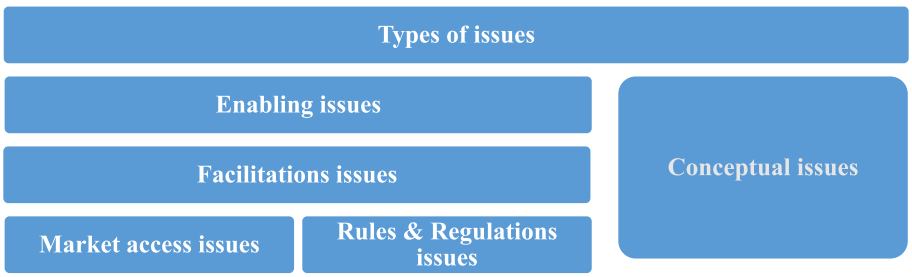


Fig. 2 Issues to consider by the AfCFTA digital trade protocol

enabling issues is also important and includes facilitation measures (paperless trade administration, cooperation, transparency, and electronic authentication) and covering issues such as market access (customs duties, digital products, nondiscrimination for electronic and digital products, and cross-border flows) and rules and regulations (protection, protection of personal information, electronic supply of information, and domestic electronic transaction frameworks).

2.3 Digital trade regulations as a key factor impacting regional digital trade: what does the literature tell us?

Although the concept of digital trade regulations is not clearly defined in the literature, it is generally apprehended by legislations, trade rules, trade agreements, standards and principles and any policy that govern digital trade (Joubert 2021). Digital trade regulations are generally related to: consumer protection, content access**,** cybersecurity, data protection**,** electronic transactions, Intellectual Property Rights (IPR), internet intermediary liability, Foreign Direct Investment (FDI), personal privacy, public procurement of ICT products and, tariffs and trade defence measures (Nemoto and López González 2021; United Nations 2021). The United Nations (2023) group the digital trade regulations into three broad clusters:

- The Traditional trade policy cluster covers regulations such as tariff and nontariff measures (NTMs) on ICT goods and services;
- The domestic regulations cluster which includes regulations on public procurement, FDI, IPR and telecommunication and competition;

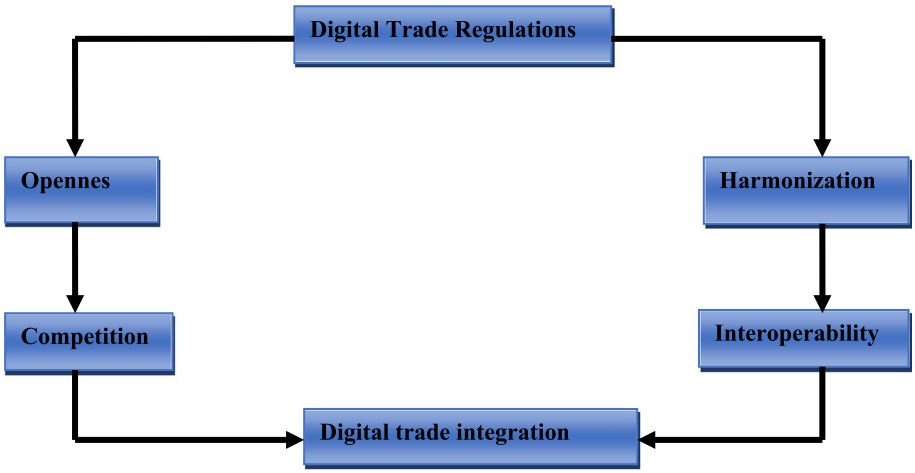


Fig. 3 Digital trade regulations and their impact on digital trade integration

- The digital governance cluster encompasses regulations that focus on cross-border data policies, domestic data protection and privacy, internet intermediary liability, content access and, online sales and transactions.

Theoretically, the impact of digital trade regulations on digital trade integration depends of their level of openness and harmonization. As Fig. 3 shows, opened and harmonized, digital trade regulations play a positive role in building the digital trade and faster digital trade integration. Indeed, the open digital trade market reduce barriers on cross-border digital trade and promotes the exchange of digital goods and services in the region. An open market reduces uncertainty, promotes transparency and stimulates investments in digital infrastructure and skills. Which therefore encourages innovations and strengthens the competitiveness of businesses (World Bank 2016). On the other hand, harmonization of the regulations of different countries avoids regulatory fragmentation. Countries that harmonized their digital trade regulations reduce trade costs between them, increases interoperability and are more digitally integrated (Daza Jaller et al.2020; United Nations 2021). In this perspective, López González and Ferencz (2018) argue that the ability to pay for the e-book will depend on the presence of interoperable e-payment methods.

When regional regulations are harmonized, countries can more easily participate in the integrated digital economy. For example, the harmonization of digital trade governance measures between many countries strengthens the security of transactions and the trust of consumers, business owners and investors and, promote digital trade (Monteiro and Teh 2017; Jaller et al.2020; Burri and Polanco 2020; Nemoto and López González 2021). Conversely, a complex, ambiguous and heterogeneous digital regulatory environment can increase the costs of doing business and digital trade integration (Monteiro and Teh 2017; Daza Jaller et al. 2020; Burri and Polanco 2020; Nemoto and López González 2021).

Empirically, several studies have confirmed the positive effects of digital trade provisions on digital trade integration. This is the case in APEC[9] region where the results of the study show that digital trade provisions increase digital trade flows. In particular, the adoption of specific digital trade provisions increases the flows of digitally ordered and digitally deliverable trade by between 11 and 44% in successive years (APEC 2023). According to the study of Nattrass and de Fazekas (2018), the open and harmonized digital trade provisions contain in the United States-Mexico-Canada Agreement (USMCA), the successor to NAFTA will positively affect digital trade in the region.

3 Digital trade: what does data shows?

The present section is divided into two stages. The first is the assessment of digital trade and digital trade regulatory ecosystem in Africa trough descriptive analysis. The second is the econometrics analysis to understand the effects of the regulatory framework on digital trade development. We used two main databases. The UNCTAD digital economy database for digital trade and the Regional Digital Trade Integration Index (RDTII) for digital trade regulatory environment. Recently compute by the United Nations Economic Commission for Africa (ECA), the RDTII cover 28 African countries[10] and includes 12 pillars (or policy areas) and 65 indicators.

[9] Asia–Pacific Economic Cooperation (APEC).

[10] See the appendix for more information about the countries, pillars and the computation of RDTII.

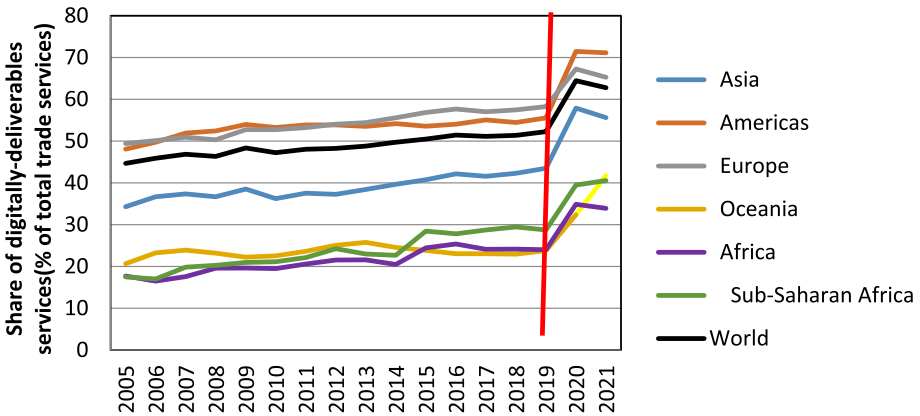


Fig. 4 Digitally deliverable trade as share of total export services (2005–2021)

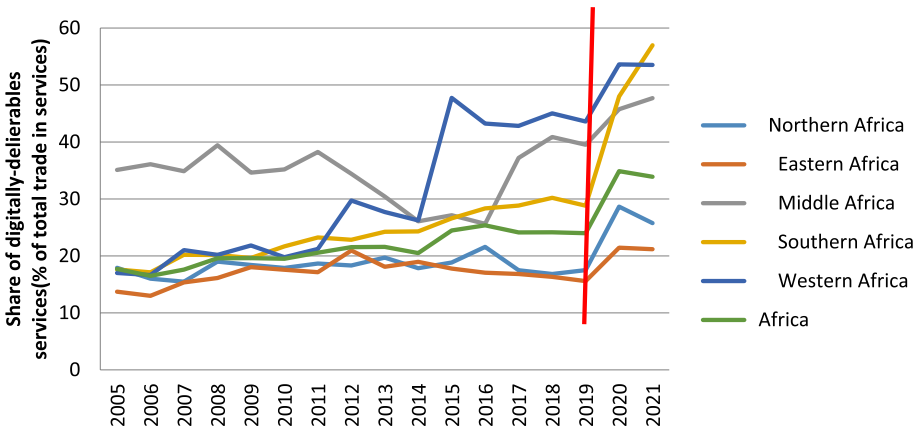


Fig. 5 Digitally deliverable trade as share of total services exports by African regions (2005–2021)

3.1 Digital trade trends and the digital trade regulations ecosystem in Africa

3.1.1 Digital trade trends in Africa

The rapid descriptive analysis shows that digital trade has increased across in all regions in the world and particularly in Africa since the beginning of the covid-19 pandemic in 2019. As a share of digitally deliverable services[11] in total export services, digital trade approached, respectively, 35%, 68% and 72% of total export services in Africa, Europe and North America (Fig. 4). However, the digitally deliverable services growth in Africa is inferior to world average over the whole period.

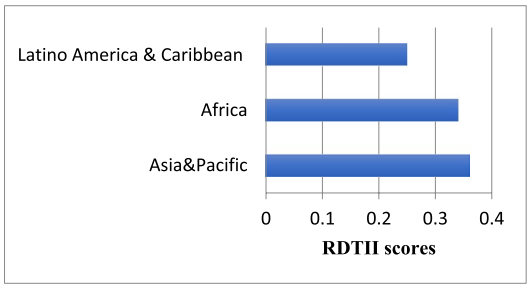
In Africa, digitally deliverable trade varied widely across sub-regions (Fig. 5). While countries in Western Africa, Southern Africa and Middle Africa saw the strongest growth of any sub-region in Africa in 2020, Northern and Eastern sub-region perform below the African average (35%). Except in Southern and Middle Africa, digitally deliverable services trade declined in 2020.

However, the growth rate of digital trade varies among African countries. One of the main factors explaining the divergence of digitally trade growth between the African countries is attributed to the digital trade regulatory environment.

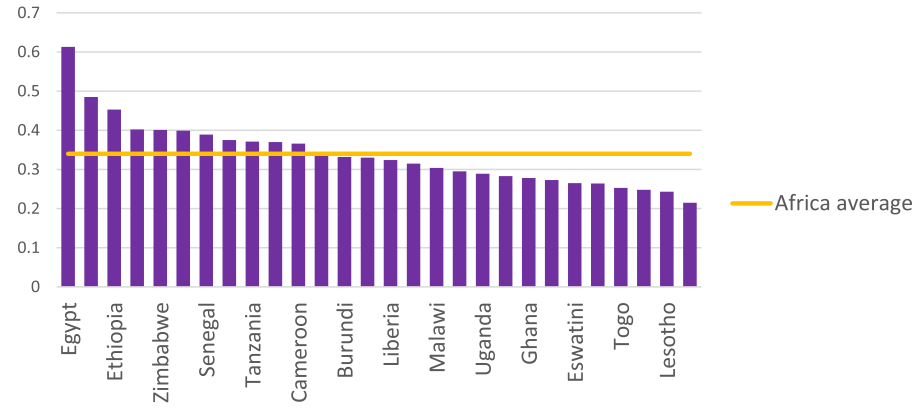
3.1.2 Digital trade regulations ecosystem in Africa

The analysis of the digital trade regulation ecosystem in Africa reveals on average, the RDTII score is 0,34 meaning that Africa is not overly restrictive. But Africa performed more than Asia and pacific (0,36) and least than Latino America and the Caribbean (0,25) (Fig. 6a). This global RDTII score in Africa hides important disparities across African countries. The RDTII scores range from

[11] Digitally-deliverable services include: insurance and pension services, financial services, charges for the use of intellectual property, telecommunications, computer and information services, other business services and audiovisual and related services.



a. RDTII scores across regions



b. RDTII scores across African countries

Fig. 6 RDTII scores across regions and African countries

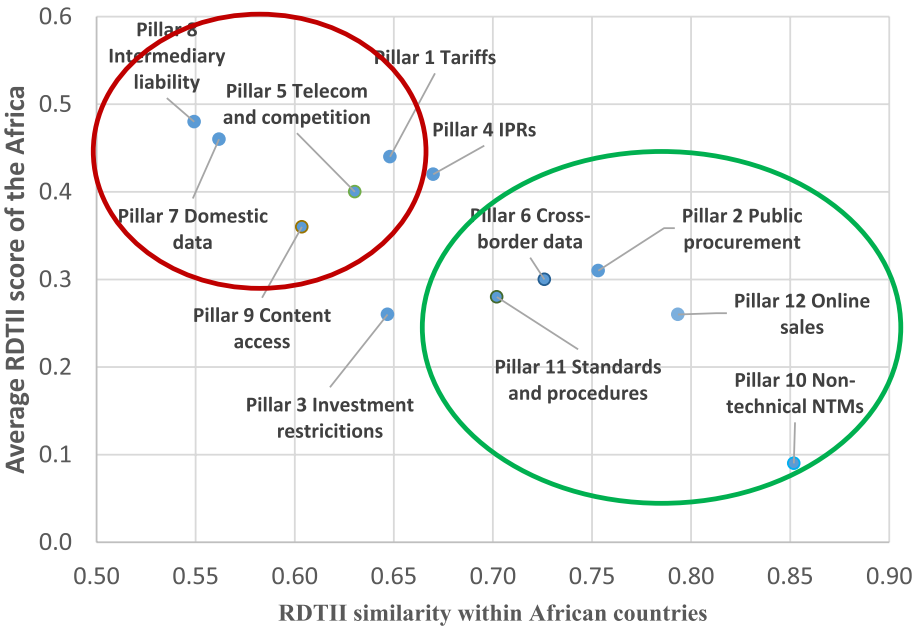


Fig. 7 Digital-trade policy diversity in the African region in 2022, based on RDTII pillars

0.22 for Botswana (the lowest) to 0.61 for Egypt (the highest), as illustrated in Fig. 6b.

However, at pillar level, the analyze of the RDTII score reveals five main impediments to digital trade integration in Africa (Fig. 7). These impediments refer to regulations on:

- Tariffs applied on ICT goods imported within the African continent;
- Intellectual Property Rights (IPRs);
- Telecom and competition;
- Domestic data policies;
- Internet intermediary liability.

As Fig. 7 shows, trade of ICT goods is subject to heavy regulations on tariffs in many African countries. Eliminating or lowering tariffs may impact positively the exchange of ICT goods between African countries and boost the AfCFTA.

The IPRs regulations constitute another impediment of the development of digital trade in Africa. Indeed, IPRs regulations are primary determinants for digital trade investment. Legislation on intellectual property rights ensures the protection of trademarks, trade secrets, copyrights and other rights. Another regulation which affects negatively digital trade integration in Africa are regulations on telecom competition. In many African countries, the State plays an important role in telecommunication sector. The mobile telecom sector is characterized by the presence of a dominant actor, which denotes the lack of competition rules. Licensing requirements are often associated with discriminatory restrictions hamper foreign telecom services providers from bidding.

The regulations governing the domestic data also impede digital trade integration in Africa. Indeed, data policies are vital to ensure the safety of personal and business data. Investors and consumers feel more confident about investing in or using digital services when the safety of data is guaranteed by law. But in many African countries, regulations on cross-border transfer of personal and non-personal data don't exist. For example, according to the UNCTAD data,[12] only 20 African countries have legislation on online consumer protection, 33 on data protection and privacy; and 39 on Cybercrime. The African Union Convention on Cybersecurity and Personal Data Protection (2014) sets out standards, principles and actions to be taken by AU member States in the areas of e-commerce, electronic contracts, personal data protection, electronic advertising, and security of electronic transactions. However, the Convention is not immediately binding. It requires ratification by fifteen countries to become a binding and actionable instrument. As at october 2023, there were only fifteen ratifications.

The intermediary liability protection is the core foundation of digital trade policies. Generally referred to Internet Service Providers (ISPs), intermediaries play an important role in our daily use of the Internet. Intermediary liability[13] occurs where governments can hold ISPs and websites liable for unlawful or harmful content created by users of those services. Fig. 7 shows that internet intermediary liability pillar score is the highest of all pillar scores. The main reason is the lack of limitations on intermediary liability (or safe harbour) in many African countries.

3.2 Econometrics analysis

3.2.1 Specification of gravity model and data

In addition to providing descriptive statistics showing the pattern of digital trade and digital trade regulatory ecosystem in Africa, we conduct an econometric analysis that allows us to investigate the effects of a battery of digital trade regulatory indicators on digital trade. Our gravity model is adapted from Anderson and Van Wincoop (2003), and specified as follows:

$$logdigitaltrade_{ij} = logY_i + logY_j + logY + (1-\sigma)\left[log\tau_{ij} - log\pi_i - logP_j\right] \tag{1}$$

$$log\pi_{ij} = a_1 logdistance_{ij} + a_2 contiguity_{ij} + a_3 comlangoff_{ij} + a_4 comcol_{ij} + a_5 rta_{ij} \tag{2}$$

$$\pi_i = \sum_{j=1}^{c} \left(\frac{\tau_{ij}}{P_j}\right)^{1-\sigma} \frac{Y_i}{Y} \tag{3}$$

[12] https://unctad.org/page/online-consumer-protection-legislation-worldwide.

[13] Center for Democracy and Technology (2010).
Intermediary liability occurs in a vast array of circumstances, around a multitude of issues including: copy right infringements, digital piracy, trademark disputes, network management, spamming and phishing, "cybercrime", defamation, hate speech, child pornography, "illegal content", offensive but legal content, censorship, broadcasting and telecommunications laws and regulations, and privacy protection.

Table 2 Summary of statistics

Variables	Observations	Mean	Standard deviation	Minimum	Maximum
Time-variant bilateral variables					
Digital trade	241	1.927543	3.061669	-6.907755	7.997947
RDTII	241	0.340303	0.064116	0.228144	0.613
Time-invariant bilateral variables					
Distance	241	7.720511	0.8304759	5.048171	9.014665
Contiguity	241	0 .1286307	0.3354875	0.00	1.00
Common currency	241	0 .1037344	0.3055503	0.00	1.00
Common Official Language	241	0.5394191	0.4994811	0.00	1.00
Common colonizer	241	0.2365145	0.425826	0.00	1.00
RTA	230	0..2608696	0.4400666	0.00	1.00

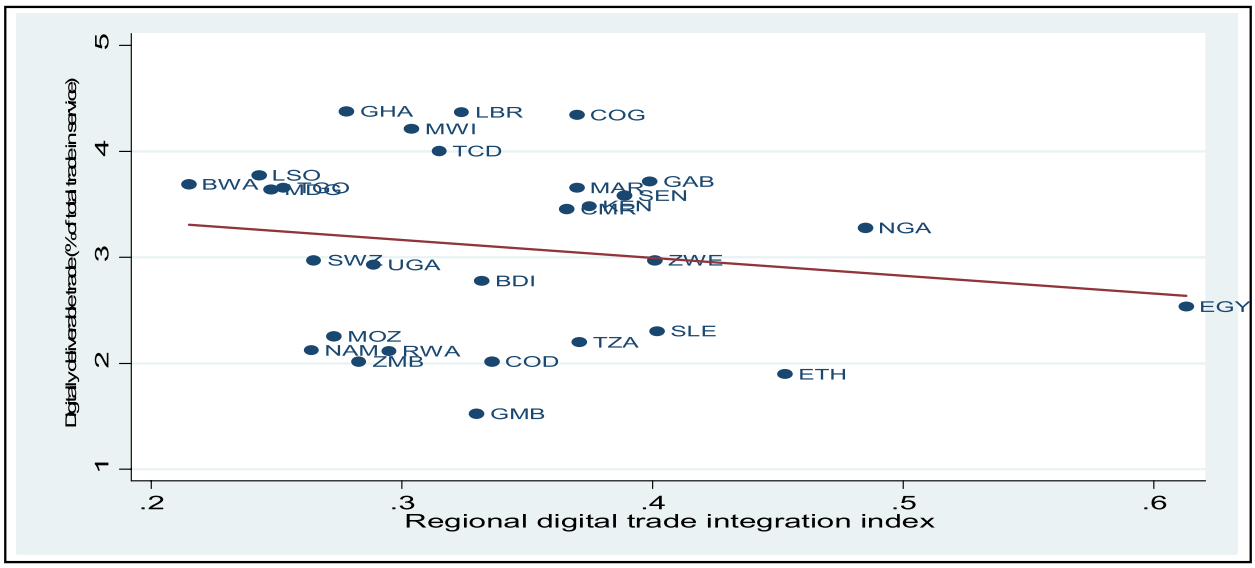


Fig. 8 Digital-trade policy and digitally deliverable trade

$$P_j = \sum_{j=1}^{c} \left(\frac{\tau_{ij}}{\pi_i} \right)^{1-\sigma} \frac{Y_j}{Y} \tag{4}$$

where $logdigitaltrade_{ij}$ is digital trade exports of country i to country j, Y_i, Y_j and Y are the exporter, importer and world GDP respectively, σ is the intra-sectoral elasticity of substitution, τ_{ij} is trade costs, $distance_{ij}$ is the geographical distance between economies i and j, $contiguity_{ij}$ is a dummy variable equal to unity if countries i and j share a common land border, $comlangoff_{ij}$ is a dummy variable equal to unity if economies i and j share a common official language, $comcol_{ij}$, and rta_{ij} is a dummy variable equal to unity if there are any free trade agreements between economies i and j. π_i represents outward multilateral resistance, which captures the fact that exports from economy i to economy j depend on trade costs across all possible export markets. P_j represents inward multilateral resistance, which captures the dependence of imports into economy i from economy j on trade costs across all possible suppliers.

After transformations and following APEC (2023), our Ordinary Least Squares (OLS) equation is:

$$logdigitaltrade_{ij} = a_0 + a_1 logdistance_{ij} + a_2 contiguity_{ij} + a_3 comlangoff_{ij} + a_4 comcol_{ij} + a_5 rta_{ij} + a_6 + logRDTII_{ij} + \varepsilon_{ij}$$

where: $RDTII_{ij}$ is the bilateral Regional Digital Trade Integration Index specific to the bilateral pair ij as a proxy

Table 3 Gravity model estimated results

	(1)	(2)	(3)	(4)
Distance	-0.75	-0.70	-0.74	-0.72
	(0.24)***	(0.30)*	(0.38)*	(0.37)*
Contiguity	-0.630	-0.362	-0.342	-0.295
	(0.75361)	(0.72237)	(0.83230)	(0.78150)
Common colonizer	0.184	0.549	0.455	0.460
	(0.580)*	(0.563)**	(0.567)*	(0.541)*
Common official language	0.693	0.806	0.659	0.741
	(0.495)	(0.505)	(0.482)	(0.464)
common currency		1.800	2.070	1.789
		(0.597)***	(0.609)***	(0.592)***
RTA			0.40	0.582
			(0.567)*	(0.568)
RDTII				-1.890
				(1.071)*
Constant	7.684	4.577	7.765	9.613
	(1.874)***	(2.363)*	(3.133)**	(3.309)***
Fixe effects	No	No	No	Yes
Adj R-squared	0.1	0.1	0.1	0.1
Observtions	241.00	241.00	230.00	230.00

All non-dummy variables are in logs. The notations * $p<0.1$; ** $p<0.05$; *** $p<0.01$ denote. significance at 10, 5 and 1 percent levels, respectively. Standard deviation in parentheses

Source: Authors' calculations

for digital trade regulatory environment and ε_{ij} is the error term. Full details of variables and data sources are in Table 1.

3.2.2 Results and discussion

Table 2 below presents the summary statistics of all variables.

The correlation analysis shows that digital trade and digital trade regulatory barriers (RDTII) are strongly negatively correlated in Africa (The Fig. 8). Countries like Botswana, Lesotho and Ghana appear to have more favorable digital trade regulatory environment which results in good performance on digital trade. In contrary, a country like Egypt pays consequences of a bad digital trade regulatory environment.

This finding supports our basic intuition that a higher RDTII (or bad digital trade regulatory environment) affects negatively intra-African digital trade. The econometric results seem to confirm this finding as shown in Table 3. Column (1) reflects a baseline model that includes the conventional gravity covariates. The geographical distance is positive and statistically significant. The estimated coefficient is consistent with prior empirical findings which is included between 0.7 and 1.5 (WTO, 2012).

In columns (2) and (3), we introduce policy variables, common currency and RTA variables respectively. Not surprisingly, this two variables have expected sign and are significant. Sharing the same currency and be a members of an RTA increase digital trade.

Column (4) finds negative and significant effects for the RDTII on digital trade with the estimate suggesting that each one-point increase of RDTII (in other words, most restricted regulatory framework) reduce digital trade by 189 percent.

4 Conclusion

As the proportion of digital trade in global trade continues to increase, the role of digital trade regulations is becoming more prominent. The objective of this study was to show in the AfCFTA context, that a conducive and harmonized regulatory environment is needed to boost intra-African digital trade. For this aim, we used both descriptive and econometrics analysis. The digital trade regulation is measure by the Regional Digital Trade Integration Index (RDTII) recently computed by the United Nations Economic Commission for Africa (ECA). The RDTII cover 28 African countries and includes 12 pillars (or policy areas). Indicators and pillars' scores range from 0 (policy environment is conducive for digital trade) to 1 (policy framework can hamper digital trade).

Our descriptive analysis shows that digital trade has increased across all regions in the world particularly in Africa since the beginning of the covid-19 pandemic. As a share of digitally deliverable services in total services exports, digital trade represents 35% of total services exports in Africa. However, digital trade expansion faces regulatory barriers and the lack of a harmonized regulatory framework. Our econometrics results confirm that digital trade and digital trade regulatory barriers are strongly negatively correlated in Africa. For example, each one-point increase of RDTII (most restricted regulatory framework) reduce digital trade by 189 percent. This result establishes the need to improve digital trade regulations framework in order to boost intra-African digital trade. In this regard, the Protocol on digital trade under negotiations must establish a conducive regulatory environment by putting in place a clear and predictable harmonized legislations, trade rules, common principles and standards for digital trade. Particular attention should be paid to regulations on domestic data, internet intermediary liability, IPRs, Tariffs applied on ICT goods imported and, telecommunication and competition. Indeed, to boost digital trade integration in the AfCFTA

context, the Protocol on digital trade must prioritize regulations on:

- domestic data that contribute to enhance trust in digital transaction. In this perspective, the adoption of the African Union Convention on Cybersecurity and Personal Data Protection (2014) is an important step for harmonization;
- internet intermediary liability. These regulations must avoid to discourage the participation of intermediary in digital trade with onerous liability and obligations;
- IPRs which are determinant for digital trade investment and innovation. The adoption of the international instruments like UNCITRAL Model Law on Electronic Signature and the WIPO Copyright Treaty can facilitate harmonization;
- tariffs applied on ICT goods that limit trade in ICT goods between African counties;
- Telecommunication and competition.

Appendix

The Regional Digital Trade Integration Index (RDTII)

The RDTII framework provides an overview of the digital trade policy environment. The index scores indicate how significantly the regulatory environment adds to "the cost of doing digital trade-related business". (United Nations 2023). The scores range from **0** (policy environment is conducive for digital trade) to **1** (policy framework can hamper digital trade).

The methodology of the RDTII is based on 12 thematic pillars. These 12 pillars, in turn, cover 48 sub-pillars in total. Calculating a RDTII score of an economy requires three steps: (1) apply a score for each sub-pillar; (2) assign a weight for each sub-pillar within a pillar; (3) give an equal weight for each pillar and calculate the average of the scores from the pillars to assign the RDTII score to the economy.

Each pillar has the same weight (simple average). But, within each pillar, entries have diverse weights to reflect the expected impact of the measure on digital trade (see United Nations (2022), for more informations about the weight within each pilar).

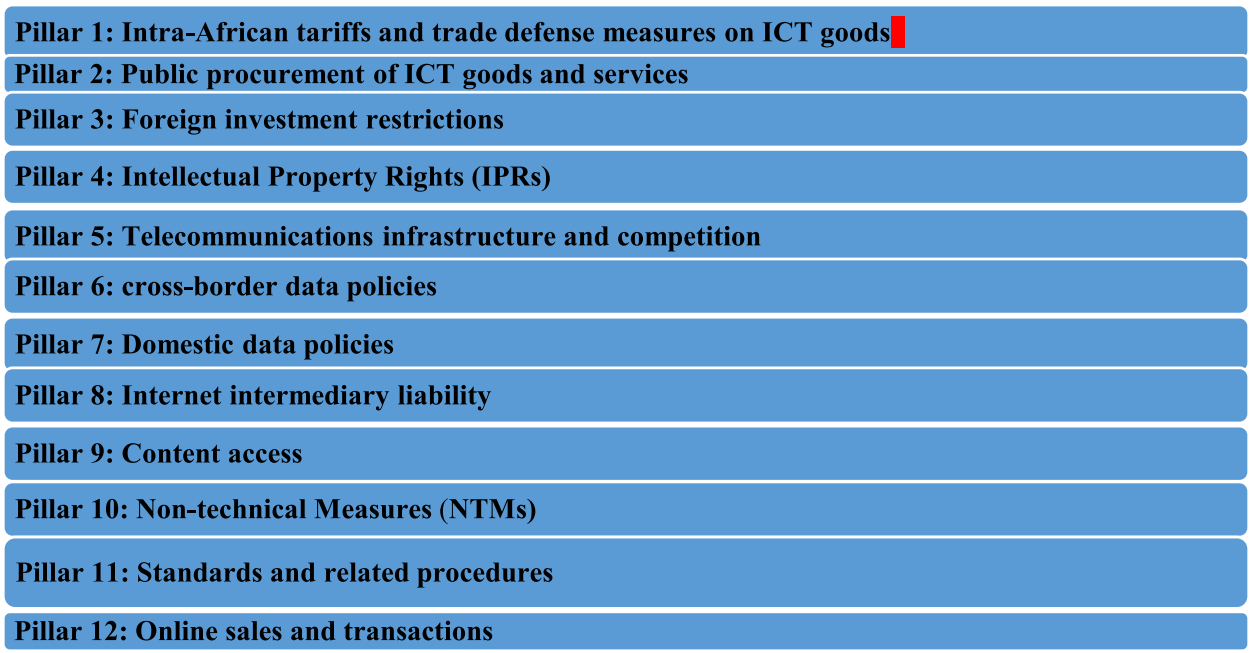


Fig. 9 Pillars cover by the RDTII

Countries cover by the RDTII

Phase 1 (11 countries): Cameroon, Chad, Gabon, Ghana, Kenya, Malawi, Nigeria, Tanzania, Uganda, Zambia, and Zimbabwe;

Phase 2 (17 countries): Burundi, Botswana, Congo, Democratic Republic of the Congo, Egypt, Eswatini, Ethiopia, Gambia, Lesotho, Liberia, Madagascar, Mozambique, Namibia, Rwanda, Senegal, Sierra Leone, and Togo.

Table 4 Countries cover by the RDTII (by regions)

Regions	Countries
Central Africa (5)	Cameroon, Chad, Congo, DRC, Gabon
Eastern Africa (7)	Burundi, Ethiopia, Kenya, Madagascar, Rwanda, Tanzania, Uganda
Western Africa (7)	Gambia, Ghana, Liberia, Nigeria, Senegal, Sierra Leone, Togo
Southern Africa (8)	Botswana, Eswatini, Lesotho, Malawi, Mozambique, Namibia, Zambia, Zimbabwe
Northern Africa (1)	Egypt

Source: author

Table 5 Some macro-economic indicators

Country	Year	Current account balance (% of GDP)	Employers, total (% of total employment)	Exports of goods and services (% of GDP)	GDP per capita (constant 2015 US$)	Gross national expenditure (% of GDP)	Population
Botswana	2000	9,41,805	2,734,174	51,83,181	4917,113	87,77,586	1,726,985
Botswana	2010	-5,8884	2,585,485	42,14,433	5428,164	108,2514	2,091,664
Botswana	2020	-8,63,616	2,261,203	31,29,028	5810,757	117,4137	2,546,402
Burundi	2000	-5,76,259	1,154,184	6,318,971	310,0067	109,9158	6,307,659
Burundi	2010	-14,8111	1,178,096	9,700,002	297,7917	120,1	9,126,605
Burundi	2020	.	1,194,965	4,54,932	263,361	117,6747	12,220,227
Cameroon	2000	-2,0669	1,944,263	24,85,681	1137,625	98,21,432	15,091,594
Cameroon	2010	-3,11,602	2,698,774	22,36,889	1277,266	102,5008	19,878,036
Cameroon	2020	-3,70,931	2,89,445	15,23,584	1419,328	103,2673	26,491,087
Chad	2000	.	0,372,605	16,8901	379,3515	117,791	8,259,137
Chad	2010	.	0,607,976	36,84,561	727,5249	106,6995	11,894,727
Chad	2020	.	1,233,806	26,71,615	622,2785	115,2981	16,644,701
Congo, Dem. Rep	2000	.	1,350,206	11,44,961	339,9694	104,1152	48,616,317
Congo, Dem. Rep	2010	-10,0785	1,839,013	41,11,115	393,0275	108,5253	66,391,257
Congo, Dem. Rep	2020	-2,15,968	2,307,309	28,60,495	487,4331	101,2819	92,853,164
Congo, Rep	2000	20,12,894	1,124,244	80,29,715	2057,767	57,69,854	3,134,030
Congo, Rep	2010	6,854,136	1,087262	68,52,809	2333,074	87,00857	4,437,884
Congo, Rep	2020	13,74,235	0,977,248	57,01271	1586,318	85,39,681	5,702,174
Egypt, Arab Rep	2000	-0,97,257	17,12,297	16,20,112	2517,581	106,6157	71,371,371
Egypt, Arab Rep	2010	-2,05668	14,74,417	21,34,925	3305,883	105,2379	87,252,413
Egypt, Arab Rep	2020	-3,70,904	8,806,324	12,47,278	3835,784	107,1807	1,07E + 08

Country	Year	Current account balance (% of GDP)	Employers, total (% of total employment)	Exports of goods and services (% of GDP)	GDP per capita (constant 2015 US$)	Gross national expenditure (% of GDP)	Population
Eswatini	2000	-2,63,874	1,521,037	71,26,906	2411,445	110,0728	1,030,496
Eswatini	2010	-8,74,814	1,959,092	46,47,295	3199,054	110,536	1,099,920
Eswatini	2020	6,803,764	1,972,733	44,91,653	3672,653	100,6715	1,180,655
Ethiopia	2000	0,163,269	0,816,013	.	258,8705	.	67,031,867
Ethiopia	2010	-2,12,226	0,565,469	.	445,1823	.	89,237,791
Ethiopia	2020	-2,52,541	0,483,776	7,127,201	811,2555	109,7517	1,17E + 08
Gabon	2000	19,70,276	1,127,385	69,03218	7611,836	63,63,758	1,272,935
Gabon	2010	17,06851	2,549,794	57,66,349	6514,731	73,83,068	1,711,105
Gabon	2020	.	2,460,089	47,50,476	6619,689	75,05047	2,292,573
Gambia, The	2000	.	1,048404	25,79,867	667,9654	105,666	1,437,539
Gambia, The	2010	1,133,912	1,333,217	14,66,781	697,1626	117,324	1,937,275
Gambia, The	2020	-4,77,622	1,682,392	9,621,792	655,5761	129,2031	2,573,995
Ghana	2000	-7,75,495	5,266,019	48,80,226	1020,162	118,4439	19,665,502
Ghana	2010	-8,53,274	4,911,401	29,47,672	1374,985	113,7834	25,574,719
Ghana	2020	-3,04665	5,176,946	20,72,105	1951,065	97,07477	32,180,401
Kenya	2000	-1,56,935	6,177,298	21,58,757	1187,326	110,7592	30,851,606
Kenya	2010	-5,21,669	1,777,603	20,12,436	1342,769	107,0427	41,517,895
Kenya	2020	-4,76,039	1,476,612	9,6404	1616,502	108,5686	51,985,780
Malawi	2000	-2,89,662	1,119,435	.	421,8195	.	11,229,387
Malawi	2010	-9,57,008	1,163,033	.	511,8183	.	14,718,422
Malawi	2020	-12,1625	1,399,646	.	562,783	.	19,377,061
Mozambique	2000	.	3,602,233	11,1962	293,7089	123,7108	17,768,505
Mozambique	2010	-15,1238	3,2151	28,44,395	489,407	113,7528	23,073,723
Mozambique	2020	-27,5507	2,444,473	29,322	576,2235	142,8839	31,178,239
Madagascar	2000	-5,62,676	1,384,344	19,11,967	474,139	104,5378	16,216,431
Madagascar	2010	-8,94,854	2,169,872	21,87,438	456,7292	114,1261	21,731,053
Madagascar	2020	-4,77,456	4,528,303	20,14,207	433,8382	113,4179	28,225,177
Namibia	2000	4,883,022	5,049077	40,73,615	3082,759	101,0145	1,819,141
Namibia	2010	-3,92,245	4,840,745	48,28,195	4186,042	110,0814	2,099,271
Namibia	2020	2,55,798	6,320,007	33,67,787	4152,181	109,5692	2,489,098
Nigeria	2000	10,69,427	1,879,983	36,02327	1462,536	76,94,907	1,23E + 08
Nigeria	2010	3,572,647	2,075573	25,66,061	2397,324	93,56,894	1,61E + 08
Nigeria	2020	-3,69,874	1,929,295	8,118,312	2401,184	100,1156	2,08E + 08
Liberia	2000	.	2,261,032	.	763,4269	.	2,895,224
Liberia	2010	-42,7136	2,017064	.	627,8054	.	4,019,956
Liberia	2020	.	1,518,473	.	612,3842	.	5,087,584
Rwanda	2000	.	0,051418	5,377,399	337,6319	116,7311	8,109,989
Rwanda	2010	-6,51,694	0,05386	10,74,289	583,8612	115,7935	10,309,031
Rwanda	2020	-12,0665	0,050419	19,32,872	821,7851	116,5384	13,146,362
Sierra Leone	2000	-17,6583	2,050159	18,13,484	400,5989	121,2607	4,584,067
Sierra Leone	2010	-22,7021	2,489,758	16,7941	537,379	117,6867	6,436,698
Sierra Leone	2020	-7,88,626	2,837,317	15,09107	608,3381	121,3758	8,233,970
Lesotho	2000	-8,58,525	2,996,327	.	727,2328	.	1,998,630
Lesotho	2010	-7,08448	3,1162	41,13,816	982,2051	165,8516	2,022,747
Lesotho	2020	-0,85,995	2,809,147	41,84,474	969,396	157,7995	2,254,100
Senegal	2000	-5,5408	0,78,677	17,76,415	1076,892	113,1518	9,704,287
Senegal	2010	-3,65,651	0,807,106	19,98,124	1163,228	112,4954	12,530,121
Senegal	2020	.	1,862,292	20,70,055	1391,047	118,6458	16,436,120
Tanzania	2000	-3,20,202	1,654,553	10,80,909	551,6566	102,3627	34,463,704

Country	Year	Current account balance (% of GDP)	Employers, total (% of total employment)	Exports of goods and services (% of GDP)	GDP per capita (constant 2015 US$)	Gross national expenditure (% of GDP)	Population
Tanzania	2010	-6,90,561	1,675,019	19,60,753	794,4539	108,4254	45,110,527
Tanzania	2020	-1,6982	3,038276	13,50,387	1028,146	101,0063	61,704,518
Uganda	2000	-5,79,341	0,267,037	10,65,141	508,762	111,4462	24,020,697
Uganda	2010	-6,03512	2,426,835	13,82,448	769,7904	111,8606	32,341,728
Uganda	2020	-9,4453	4,156,848	15,4148	918,1201	104,9794	44,404,611
Zambia	2000	-18,3981	0,400,017	23,92,386	814,773	.	9,891,136
Zambia	2010	7,525,491	0,286,043	37,02593	1198,305	93,84,893	13,792,086
Zambia	2020	5,310,805	0,278,853	46,79,017	1237,283	85,48,638	18,927,715
Zimbabwe	2000	.	0,333,065	38,15,975	1676,515	97,7479	11,834,676
Zimbabwe	2010	-13,7472	0,587,737	29,6409	1096,835	123,8424	12,839,771
Zimbabwe	2020	5,096583	0,312,304	25,91,701	1213,117	102,8165	15,669,666
Togo	2000	-6,90,503	1,189,334	20,7646	675,465	106,1147	5,008,035
Togo	2010	-4,28,941	1,559,131	27,84,441	641,6734	116,2315	6,571,855
Togo	2020	-0,28,064	1,598,488	23,70,187	833,8599	108,7887	8,442,580
East Asia & Pacific	2000	.	3,91,606	23,62,741	5015,23	.	2,05E + 09
East Asia & Pacific	2010	.	4,010055	32,06071	7771,428	.	2,21E + 09
East Asia & Pacific	2020	.	3,532,474	26,68,369	11,130,37	.	2,36E + 09
Europe & Central Asia (excluding high income)	2000	.	2,562,719	34,24,185	4084,46	.	3,71E + 08
Europe & Central Asia (excluding high income)	2010	.	2,983,277	30,0287	6508,074	.	3,79E + 08
Europe & Central Asia (excluding high income)	2020	.	3,116,988	29,38,308	8018,716	.	4,01E + 08
Sub-Saharan Africa	2000	.	2,212,815	29,27,995	1238,058	.	6,71E + 08
Sub-Saharan Africa	2010	.	2,0322	29,29,306	1589,108	.	8,8E + 08
Sub-Saharan Africa	2020	.	2,286,849	19,10,286	1577,167	.	1,15E + 09
World	2000	.	3,288,256	23,57,926	7871,168	.	6,14E + 09
World	2010	.	3,278,881	28,74,478	9309,046	.	6,97E + 09
World	2020	.	3,199,557	26,38,327	10,499,65	.	7,82E + 09

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KOUTY Manfred wrote, read and appoved the final draft of the manuscript.

Declarations

Competing interests
The authors declare they have no conflicts of interest.

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